

# Machine Learning for Solid-State Battery Materials Discovery: A Comprehensive Survey

## Abstract

The transition to solid-state batteries (SSBs) represents a paradigm shift in energy storage, promising enhanced safety and energy density through the replacement of flammable liquid electrolytes with solid ionic conductors. However, the discovery and optimization of solid-state electrolytes (SSEs), stable interfaces, and compatible electrode materials remain bottlenecked by the vastness of chemical space and the complexity of multiscale physicochemical phenomena. This survey provides a comprehensive review of the rapidly evolving landscape where machine learning (ML) and artificial intelligence (AI) intersect with solid-state battery materials science. We synthesize findings from 300 recent studies to construct a detailed taxonomy of methodological approaches, ranging from high-throughput screening and universal machine-learned interatomic potentials (uMLIPs) to generative design frameworks and autonomous self-driving laboratories. We critically analyze the efficacy of graph neural networks, equivariant architectures, and large language models in predicting ionic conductivity, electrochemical stability, and mechanical properties. Furthermore, we examine the integration of ML with first-principles calculations to resolve interfacial degradation mechanisms, dendrite growth, and chemomechanical failure modes. By evaluating current benchmarks, dataset limitations, and the transition from static property prediction to dynamic process optimization, this paper outlines the critical challenges and future trajectories for AI-driven acceleration of solid-state battery commercialization.

## 1. Introduction

The global imperative for decarbonization has intensified the search for next-generation energy storage systems capable of surpassing the theoretical limits of conventional lithium-ion batteries. Solid-state batteries (SSBs), which utilize solid electrolytes instead of liquid organic solvents, offer a compelling solution by mitigating flammability risks and enabling the use of high-capacity lithium metal anodes [1], [2]. Despite these advantages, the practical deployment of SSBs is hindered by significant materials challenges, including low ionic conductivity at room temperature, unstable electrode-electrolyte interfaces, and mechanical degradation during cycling [3], [4]. The traditional trial-and-error approach to materials discovery is insufficient to navigate the combinatorial complexity of multicomponent chemistries and processing conditions required to overcome these barriers.

In recent years, machine learning (ML) has emerged as a transformative tool in materials science, enabling the rapid screening of vast chemical spaces and the elucidation of complex structure-property relationships [5], [6]. In the context of SSBs, ML methodologies have evolved from simple descriptor-based regression models to sophisticated deep learning architectures capable of processing raw crystallographic data, simulating atomistic dynamics, and even generating novel material hypotheses [7], [8]. The integration of ML with high-throughput density functional theory (DFT) and molecular dynamics (MD) simulations has created a “computational microscope,” allowing researchers to probe ion transport mechanisms, interfacial reactions, and failure modes with unprecedented resolution and speed [9], [10].

This survey aims to provide a rigorous and systematic overview of the state-of-the-art in ML applications for solid-state battery materials discovery. Unlike previous reviews that focus broadly on battery informatics, this work specifically targets the unique challenges of the solid-state do-

main, such as the prediction of grain boundary effects, the modeling of amorphous phases, and the simulation of chemomechanical coupling. We organize the field into coherent methodological families, analyzing the progression from data curation and feature engineering to advanced generative models and autonomous experimental loops. By synthesizing evidence from 300 distinct studies published between 2018 and 2026, we highlight the technical innovations driving the field, compare the performance of diverse algorithmic approaches, and identify critical gaps in current methodologies. Our analysis covers the full spectrum of the materials discovery pipeline, including the development of universal interatomic potentials for accurate dynamics [11], [12], the application of large language models for hypothesis generation [13], [14], and the deployment of active learning frameworks for optimizing synthesis protocols [15], [16].

## 2. Method

The application of machine learning to solid-state battery materials can be categorized into several interconnected methodological systems. These range from static property prediction and high-throughput screening to dynamic simulations using machine-learned potentials, generative design, and autonomous discovery loops. This section provides a detailed taxonomy and synthesis of these approaches.

### 2.1. Data Curation, Representation, and Benchmarking

The foundation of any successful ML application in materials science is the quality and representativeness of the underlying data. In the domain of solid-state electrolytes (SSEs), data scarcity and heterogeneity have historically been major bottlenecks. Recent efforts have focused on constructing curated datasets that integrate experimental measurements with computational outputs, while simultaneously developing robust strategies for data splitting and leakage prevention.

**2.1.1. Curated Datasets and Leakage Prevention** A critical advancement in data curation is the recognition of data leakage risks when splitting datasets by random entries rather than by material composition or source. The OBELiX dataset represents a significant milestone, aggregating experimental room-temperature ionic conductivities and structural data while implementing a Monte Carlo splitting algorithm that groups entries by composition and publication source to prevent leakage [17]. This approach revealed that standard graph neural networks often overfit small experimental datasets and fail to capture partial occupancy effects, which are crucial for ion transport in disordered solids [17]. Similarly, the LiPS-25 benchmark framework was established to evaluate machine-learned interatomic potentials (MLIPs) across static and dynamic tasks, curating crystalline, amorphous, and liquid Li-P-S structures to assess model transferability [18]. For migration barriers, a literature-curated dataset comprising 619 distinct DFT-NEB calculations across 443 compositions has been compiled, providing a diverse chemical space coverage unattainable by single-study screening [19].

**2.1.2. Feature Engineering and Representations** The choice of material representation significantly influences model performance. Traditional approaches rely on hand-crafted descriptors derived from elemental properties and crystallographic features. For instance, facile descriptors based on tabulated elemental and structural properties have proven sufficient for high-accuracy classification of superionic conductors, achieving 82% accuracy in cross-domain predictions between sodium and lithium systems [20]. In Na-ion conductors, explicit beta regression models utilizing only

formula-based descriptors such as Na-ion content and average metal ionic radius have successfully predicted room-temperature conductivity with an  $R^2$  of 0.718 [21].

However, the field is shifting towards end-to-end learning from raw structural data. Crystal Graph Convolutional Neural Networks (CGCNN) and their variants have become standard for mapping crystal structures to properties. A 13-layer CNN inspired by VGG16 has been employed to directly map raw Crystallographic Information Files (CIF) to ionic conductivity in NASICON materials, eliminating the need for manual feature engineering [22]. Furthermore, transformer-inspired architectures like CrabNet have demonstrated superior accuracy over tree-based models when using only compositional data for predicting gravimetric capacity and voltage [23]. To address the specific challenges of electrolyte formulations, formulation graph convolutional networks (F-GCN) process multiple SMILES strings scaled by molar percentages, effectively capturing mixture physics without explicit feature engineering [24], [25]. More recently, large language models (LLMs) have been adapted to encode crystallographic disorder as categorical text descriptors, offering a complementary approach to numerical feature engineering [8].

## 2.2. High-Throughput Screening and Property Prediction

High-throughput screening (HTS) powered by ML surrogates has become the primary engine for identifying candidate materials from millions of possibilities. These workflows typically employ a hierarchical filtering strategy, where fast ML models prune the search space before expensive DFT or MD validations.

**2.2.1. Hierarchical Screening Workflows** Large-scale screening campaigns have leveraged cloud computing and ML potentials to evaluate tens of millions of candidates. One notable workflow initiated with ionic substitution on 54 elements against ICSD prototypes to generate 32.6 million candidates, using M3GNet ML potentials for rapid geometric optimization and thermodynamic stability assessment, ultimately identifying nearly 600,000 stable materials [5]. Similarly, a hierarchical screening framework combined ML interatomic potentials (CHGNet) with DFT to filter 320,000 NASICON and garnet compositions, significantly reducing the computational cost of identifying alkaline-stable conductors [26]. For magnesium battery cathodes, a sequential multi-model screening workflow integrated modified CGCNN for voltage prediction and NequIP for force field generation, successfully filtering candidates from the Materials Project and GNoME databases [27].

**2.2.2. Predicting Ionic Conductivity and Stability** Predicting ionic conductivity remains a central challenge due to its dependence on dynamic processes. ML models have been trained to predict conductivity from static structural features with varying degrees of success. A hybrid framework combining Gradient Boosting Regressors (GBR) with LLMs achieved a Mean Absolute Error (MAE) of 0.798 log(S/cm) by encoding crystallographic disorder as text descriptors [8]. Phonon-related descriptors, specifically low-frequency density of states overlap and vibrational entropy, have been shown to significantly enhance prediction accuracy, with logistic regression models achieving 93.3% accuracy in classifying superionic conductors [28].

Beyond conductivity, electrochemical and chemical stability are critical. ML-aided platforms have integrated high-throughput screening with ensemble surrogate models to filter candidates based on stability against reduction/oxidation and electronic conductivity [29]. For air-stable chalcogenide electrolytes, bond valence mapping combined with DFT-based NEB calculations identified the Li2BMQ4 family, where substituting P5+ with softer acids significantly enhanced moisture

stability [30]. In the realm of halide electrolytes, a fine-tuned CHGNet framework resolved partial occupancy disorder in  $\text{Li}_3\text{YCl}_6\text{--xBr}_x$  systems, predicting room-temperature conductivities that matched AIMD results [31].

**2.2.3. Cross-Domain and Multi-Ion Prediction** Transfer learning and multi-ion models are addressing data scarcity in post-lithium systems. A universal CGCNN trained on multi-ion intercalation data successfully predicted voltages for magnesium compounds by pooling working ion vectors, overcoming the limitations of small Mg datasets [27]. Similarly, interpretable deep transfer learning models pre-trained on abundant Li-ion data have reduced prediction errors for multivalent systems like Zn-ion batteries by freezing convolutional layers and fine-tuning only the final layers [32]. Cross-domain classification has also been demonstrated, where models trained exclusively on sodium-based data achieved high accuracy on lithium-based compounds using simple elemental and structural descriptors [20].

## 2.3. Machine-Learned Interatomic Potentials and Dynamics

While static screening identifies promising candidates, understanding ion transport mechanisms and interfacial dynamics requires accurate and efficient simulations of atomic motion. Machine-learned interatomic potentials (MLIPs) have revolutionized this domain by bridging the accuracy gap between DFT and the efficiency of classical force fields.

**2.3.1. Universal Potentials and Fine-Tuning** The development of universal MLIPs (uMLIPs) trained on massive datasets like OMat24 and MPtrj has enabled zero-shot predictions across diverse chemistries. Benchmarking studies reveal that architectural refinements, such as those in Orb-v3, primarily benefit static barrier predictions, while the inclusion of high-temperature non-equilibrium data is crucial for accurate diffusivity predictions [11]. MatterSim has demonstrated superior accuracy in energy, forces, and diffusivity for halide electrolytes due to rigorous physical consistency and active learning data coverage [33]. However, uMLIPs often require fine-tuning for specific systems. For solid-solid battery interfaces, a replay-augmented continual fine-tuning scheme (FIRE) preserved general knowledge while achieving energy RMSEs below 1 meV/atom, an order-of-magnitude improvement over existing models [34]. Similarly, fine-tuning CHGNet on transition-state data reduced migration barrier prediction MAE from 0.24 eV to 0.07 eV, enabling high-throughput NEB calculations [35].

**2.3.2. Accelerating Molecular Dynamics** MLIPs enable molecular dynamics (MD) simulations at timescales and system sizes inaccessible to AIMD. Deep Potential (DP) models have been used to simulate Li-ion transport in  $\text{Li}_3\text{TiCl}_6$  with a 3730x speedup over AIMD, capturing anti-correlated ion motion and interstitial hopping mechanisms [36]. In sulfide electrolytes, a pre-trained DPA-SSE model trained on extensive out-of-equilibrium configurations accurately predicted dynamic trajectories and ion conductivity, validating the importance of diverse training data [12]. For amorphous systems, NequIP-based potentials trained on diverse LiPON configurations achieved DFT-level accuracy (5.5 meV/atom energy MAE) while quantifying anisotropic  $\text{Li}^+$  transport [37].

**2.3.3. Mechanistic Insights from MLMD** ML-accelerated MD has provided profound insights into ion transport mechanisms. In glassy thiophosphates, MLMD revealed that corner-sharing  $\text{PS}_4\text{--BS}_4$  motifs disrupt insulating networks, facilitating fast ion conduction [38]. The “paddle-wheel” mechanism, where polyanion rotation enhances cation mobility, has been quantified in plastic crystal-polymer composites, showing that polymers induce a high-entropy interphase

that accelerates bond rotation [39]. Furthermore, constraint algorithms in MLMD have disentangled the contributions of translation, rotation, and vibration to superionicity, revealing that one mode typically dominates per material class [40]. Path entropy frameworks, utilizing Markov State Models constructed from MLMD trajectories, have quantified the multiplicity of diffusion pathways, showing that path entropy governs kinetic processes more effectively than activation energy alone in dense ionic systems [41].

## 2.4. Generative Design and Inverse Materials Discovery

Moving beyond screening existing databases, generative AI models are being employed to design novel materials with targeted properties. These approaches utilize variational autoencoders (VAEs), diffusion models, and large language models to explore uncharted chemical spaces.

**2.4.1. Diffusion Models and Generative Flows** Diffusion models have shown remarkable capability in generating stable crystal structures. The MatterGen diffusion model, integrated with ML potential prescreening, has been used to discover inorganic electrides by restricting the search to electron-rich compositions, drastically reducing combinatorial complexity [42]. For open-tunnel oxides, a hybrid pipeline combining Crystal Diffusion Variational Autoencoder (CDVAE) with fine-tuned LLaMA-3.1 generated thermodynamically stable structures superior to those produced by LLMs alone [43]. Symmetry-aware hierarchical architectures like SHAFT decompose crystal generation into space group selection and atom-lattice configuration, enforcing symmetry constraints and achieving higher match rates than baseline GFlowNets [44].

**2.4.2. LLM-Driven Hypothesis Generation** Large Language Models are increasingly used for hypothesis generation and analogical reasoning. The LacMaterial framework employs cross-domain analogy retrieval to inspire structural redundancy in solid-state electrolytes, mapping abstract relational principles from non-material domains to battery chemistry [7]. ChatBattery orchestrates a multi-agent chain of LLMs to generate, validate, and rank hypotheses for cathode optimization, reducing hallucination rates through expert-defined constraints and retrieval mechanisms [13]. Similarly, an LLM-driven workflow has generated non-trivial synergistic hypotheses by combining mechanisms from disparate literature sources, successfully recovering known design rules for high-entropy alloys and solid electrolytes [14].

**2.4.3. Inverse Design for Formulations** Generative models are also applied to electrolyte formulations. A unified predictive and generative framework integrates forward GNN prediction with inverse conditional diffusion to design molecular mixtures with multi-condition constraints, bridging the accuracy gap between simulation and reality [45]. For organic photocatalysts and solvents, neuro-symbolic search systems decouple discrete topology generation via LLMs from continuous ratio refinement, enforcing diversity to prevent mode collapse [46].

## 2.5. Interface Engineering and Degradation Modeling

The performance and longevity of SSBs are largely dictated by the stability and properties of the interfaces between electrodes and electrolytes. ML methods are being deployed to model interfacial reactions, SEI formation, and chemomechanical degradation.

**2.5.1. Interfacial Stability and SEI Formation** Predicting interfacial stability is critical for preventing degradation. Ab initio thermodynamic screening has identified LiCl as an opti-

mal passivation layer for Li-In-Cl systems, differentiating SEI mechanisms based on the electronic conductivity of reaction products [47]. ML-accelerated reaction network analysis has constructed massive electro-chemical reaction networks (eCRN) coupling salt and solvent decomposition, discovering 28 novel SEI species validated by mass spectrometry [48]. Deep learning potentials have enabled the simulation of SEI formation at Li|Li6PS5Cl interfaces, revealing a two-step mechanism involving rapid amorphous phase formation followed by slower crystallization [49]. Graph contrastive learning has been applied to cluster Li-centered micro-environments in amorphous SEIs, identifying kinetic bottlenecks without supervision [50].

**2.5.2. Chemomechanical Failure and Dendrite Growth** Mechanical failure, including void formation and dendrite growth, is a major challenge. Phase-field models coupled with ML surrogates have simulated void evolution at Li-electrolyte interfaces, showing that creep deformation dominates void shrinkage under applied pressure [51]. A minimal power dissipation model derived analytically predicts critical current densities for dendrite growth by approximating interfacial defects as ellipsoids, integrating residual electron conductivity effects [52]. Multiscale modeling frameworks combining active learning MD with finite element homogenization have revealed that grain boundaries enhance diffusion in low-diffusivity phases but suppress it in fast-diffusing ones, explaining non-Arrhenius behavior in argyrodites [53]. Furthermore, “interfacial breathing” frameworks utilizing coupled Cahn-Hilliard/Allen-Cahn equations have identified dual-timescale memory effects as a dynamic failure law, showing that stack pressure reduces void amplitude but leaves re-active memory invariant [54].

**2.5.3. Characterization and Imaging Analysis** ML is also enhancing the analysis of experimental characterization data. Cryogenic electron microscopy (Cryo-EM) combined with ML segmentation has revealed short-range order in amorphous SEI layers, correlating structural features with high reversibility [55]. Neutron imaging frameworks like ANISSA integrate dual-mode tomography to visualize lithium distribution and structural cracks simultaneously, providing superior contrast for Li detection [56]. Graph neural networks have been applied to X-ray microscopy images to construct topology-aware microstructure graphs, encoding multiphase connectivity and interface areas for high-throughput morphological analysis [57].

## 2.6. Autonomous Discovery and Self-Driving Laboratories

The ultimate goal of AI in materials science is the automation of the discovery process. Self-driving laboratories (SDLs) integrate robotic synthesis, characterization, and AI-driven decision-making into closed-loop systems.

**2.6.1. Agentic Frameworks and Workflow Automation** Recent advancements feature multi-agent systems where specialized LLM agents coordinate scientific reasoning and tool use. The SparksMatter framework orchestrates Scientist, Planner, and Assistant agents to frame hypotheses, devise tool-use sequences, and execute code, enabling dynamic exploration of chemical space [58]. Lang2MLIP formulates MLIP development as a sequential decision problem solved by LLM agents, demonstrating emergent curriculum progression from component identification to autonomous training [59]. For air-sensitive lithium halide spinels, an abductive-inductive agentic framework integrated with robotic glovebox synthesis increased the success rate of finding conductive materials fourfold by decoupling anomaly explanation from trend extrapolation [16].

**2.6.2. Active Learning and Optimization** Active learning strategies are essential for efficient exploration. An iterative ML-driven framework utilizing Particle Swarm Optimization (PSO) selected candidate parameters for NCM cathode synthesis, incorporating suboptimal experimental data to define performance boundaries and prevent model overconfidence [15]. In electrolyte optimization, Gaussian Process Regression combined with high-throughput experimentation optimized  $\text{Li}_2\text{CO}_3$  precipitation yields by tracking elemental concentrations rather than full speciation, achieving 83% yield in two cycles [60]. Autonomous scanning electrochemical cell microscopy (SECCM) with noise-aware active learning has enabled rapid exploration of compositionally complex spaces, accurately predicting composition-activity trends after measuring only 15% of the library [61].

## 2.7. Synthesis Protocol Prediction and Optimization

Bridging the gap between computational prediction and experimental realization requires modeling synthesis protocols. ML models are increasingly used to predict synthesizability and optimize processing conditions.

**2.7.1. Protocol Representation and Extraction** Extracting synthesis information from literature is a prerequisite for data-driven synthesis planning. The SynthASSBs framework employs BiLSTM-CRF sequence taggers and rule-based relation extractors to annotate ASSB synthesis procedures as flow graphs, capturing implicit links between operations and materials [62]. Beyond extraction, generative models are being trained to predict synthesis protocols. A synthesis-first paradigm treats executable protocols as primary design variables, integrating machine-readable representations with inverse-design models to bridge the divide between prediction and laboratory reality [63].

**2.7.2. Processing-Structure-Property Relationships** Understanding the impact of processing on material properties is critical. A differential manifold descriptor approach decouples chemistry and processing effects by tracking geometric shifts in manifold space induced by processing conditions, quantifying non-linear effects without distorting chemical features [64]. For solid-state reactions, a cellular automaton model (ReactCA) captures pairwise interface reactions and spatial constraints, successfully predicting selectivity trends in  $\text{BaTiO}_3$  synthesis [65]. In the context of sintering, ultrafast high-temperature sintering (UHS) combined with ML optimization has been used to decouple densification from crystal growth in LAGP electrolytes, expanding the glass thermal stability window [66].

## 3. Challenges and Outlook

Despite significant progress, several critical challenges remain in the application of ML to solid-state battery materials. Addressing these limitations is essential for translating computational insights into commercially viable technologies.

### 3.1. Data Scarcity, Quality, and Generalization

The scarcity of high-quality experimental data, particularly for ionic conductivity and interfacial stability, remains a primary bottleneck. While datasets like OBELiX [17] and LiPS-25 [18] have improved data curation, the field still suffers from a bias towards stable, easily synthesized materials, limiting the ability of models to generalize to novel chemistries. The “synthesizability gap” persists, where computationally predicted stable materials often prove difficult or impossible

to synthesize experimentally [63]. Furthermore, the heterogeneity of data sources, with varying exchange-correlation functionals in DFT studies and inconsistent experimental protocols, introduces noise that degrades model performance [19]. Future efforts must prioritize the creation of standardized, multi-fidelity datasets that integrate computational and experimental data with rigorous uncertainty quantification.

### **3.2. Accuracy and Transferability of Interatomic Potentials**

While uMLIPs have shown promise, their accuracy in predicting dynamic properties like ion diffusivity and activation energies varies significantly across architectures and chemical spaces [11], [33]. Models often struggle with long-range electrostatic interactions and charge transfer effects, which are critical at interfaces and in polar materials [67], [68]. The computational cost of training equivariant models and the risk of catastrophic forgetting during fine-tuning also pose practical challenges [37], [18]. Developing more data-efficient training strategies and architectures that explicitly incorporate physical priors, such as long-range electrostatics and many-body interactions, is a key direction for future research.

### **3.3. Modeling Complex Interfaces and Degradation**

Modeling the complex, evolving interfaces in SSBs remains computationally demanding. Current MLMD simulations are often limited to nanosecond timescales and small system sizes, which may be insufficient to capture slow degradation processes like SEI growth or dendrite propagation [69], [70]. Additionally, most models assume idealized interfaces, neglecting the effects of surface roughness, grain boundaries, and impurities that dominate real-world behavior [71], [53]. Integrating multiscale modeling approaches that seamlessly couple atomistic simulations with continuum models, while accounting for chemomechanical coupling and stochastic defect formation, is necessary to achieve predictive fidelity [9], [72].

### **3.4. Autonomous Discovery and Experimental Validation**

The deployment of self-driving laboratories is still in its infancy, with many systems limited to specific material classes or synthesis techniques [16], [61]. The integration of diverse characterization tools and the development of robust decision-making algorithms that can handle experimental failures and unexpected results remain open problems. Moreover, the “black box” nature of many deep learning models hinders the extraction of interpretable design rules, limiting their utility for guiding human intuition [73]. Future SDLs must incorporate explainable AI components and robust safety protocols to handle hazardous materials and high-energy processes.

### **3.5. Safety and Deployment Considerations**

As AI-driven discovery accelerates, safety considerations become paramount. The generation of novel materials with unknown toxicity or reactivity profiles poses potential risks [2]. Additionally, the reliance on proprietary models and datasets, such as GNoME, limits reproducibility and community validation [6]. Ensuring the transparency of AI models, the open sharing of data, and the rigorous experimental validation of AI-predicted materials are essential for the responsible development of solid-state battery technologies.



## 4. Conclusion

Machine learning has fundamentally transformed the landscape of solid-state battery materials discovery, enabling the rapid screening of millions of candidates, the accurate simulation of atomistic dynamics, and the generation of novel material hypotheses. From hierarchical high-throughput screening workflows to universal interatomic potentials and autonomous self-driving laboratories, the methodological toolkit available to researchers has expanded dramatically. These advances have led to the identification of new superionic conductors, the elucidation of complex ion transport mechanisms, and the optimization of synthesis protocols.

However, significant challenges remain in data quality, model transferability, and the accurate modeling of complex interfacial phenomena. Bridging the gap between computational prediction and experimental realization requires a concerted effort to standardize data, improve the physical fidelity of ML models, and integrate AI more deeply into the experimental loop. As the field matures, the synergy between AI, high-performance computing, and robotic experimentation promises to accelerate the development of safe, high-energy-density solid-state batteries, bringing us closer to a sustainable energy future. The next decade will likely witness the transition from AI-assisted discovery to AI-driven design, where intelligent systems not only predict materials but also autonomously navigate the complex landscape of synthesis, processing, and device integration.

## References

- [1] L.J. Zhang. (2023). State of the Art Development on Solid-State Lithium Batteries. arXiv:2307.03620.
- [2] Vittorio Pellegrini, Silvia Bodoardo, Daniel Brandell. (2019). Challenges and perspectives for new material solutions in batteries. arXiv:2004.02114.
- [3] Denys Butenko, Mustafa Khan, Liusuo Wu. (2024). Modern Solid Electrolytes for All-Solid-State Batteries: Materials Chemistry, Structure, and Transport. arXiv:2604.17380.
- [4] Arianna Massaro, Lorenzo Squillantini, Francesca De Giorgio. (2025). Advancements in Solid-State Sodium-Based Batteries: A Comprehensive Review. arXiv:2505.04391.
- [5] Chi Chen, Dan Thien Nguyen, Shannon J. Lee. (2024). Accelerating computational materials discovery with artificial intelligence and cloud high-performance computing: from large-scale screening to experimental validation. arXiv:2401.04070.
- [6] Miao Liu, Sheng Meng. (2024). Recent Breakthrough in AI-Driven Materials Science: Tech Giants Introduce Groundbreaking Models. arXiv:2402.05799.
- [7] Hongyu Guo. (2025). LacMaterial: Large Language Models as Analogical Chemists for Materials Discovery. arXiv:2510.22312.
- [8] Haewon Kim, Taekgi Lee, Seongeun Hong. (2025). Data-driven Prediction of Ionic Conductivity in Solid-State Electrolytes with Machine Learning and Large Language Models. arXiv:2601.10997.
- [9] Musawenkosi K. Ncube, Pallab Barai, Selva Chandrasekaran Selvaraj. (2024). Ionic Interdiffusion at Cathode|Solid-Electrolyte Interface: A Machine Learning– Assisted Multiscale Investigation and Mitigation Strategies. arXiv:2511.11976.
- [10] Zhenming Xu, Yongyao Xia. (2024). Progress, challenges and perspectives of computational studies on glassy superionic conductors for solid-state batteries. arXiv:2204.05118.

- [11] Xingyu Guo, Cheng Gui, Zhenbin Wang. (2024). Are Universal Potentials Ready for Alkali-Ion Battery Kinetics?. arXiv:2601.10938.
- [12] Ruoyu Wang, Mingyu Guo, Yuxiang Gao. (2024). A Pre-trained Deep Potential Model for Sulfide Solid Electrolytes with Broad Coverage and High Accuracy. arXiv:2406.18263.
- [13] Shengchao Liu, Hannan Xu, Yan Ai. (2024). Expert-Guided LLM Reasoning for Battery Discovery: From AI-Driven Hypothesis to Synthesis and Characterization. arXiv:2507.16110.
- [14] Quanliang Liu, Maciej P Polak, So Yeon Kim. (2024). Beyond designer’s knowledge: Generating materials design hypotheses via large language models. arXiv:2409.06756.
- [15] Seon-Hwa Lee, Insoo Ye, Changhwan Lee. (2024). Accelerating Battery Material Optimization through iterative Machine Learning. arXiv:2505.18162.
- [16] Yuxing Fei, Bernardus Rendy, Xiaochen Yang. (2025). Agentic LLM Reasoning in a Self-Driving Laboratory for Air-Sensitive Lithium Halide Spinel Conductors. arXiv:2604.11957.
- [17] Felix Therrien, Jamal Abou Haibeh, Divya Sharma. (2024). OBELiX: A Curated Dataset of Crystal Structures and Experimentally Measured Ionic Conductivities for Lithium Solid-State Electrolytes. arXiv:2502.14234.
- [18] Natascia L. Fragapane, Volker L. Deringer. (2024). Li–P–S Electrolyte Materials as a Benchmark for Machine-Learned Interatomic Potentials. arXiv:2511.16569.
- [19] Reshma Devi, Avaneesh Balasubramanian, Keith T. Butler. (2024). A literature-derived dataset of migration barriers for quantifying ionic transport in battery materials. arXiv:2508.06459.
- [20] Yijie Xu, Yun Zong, Kedar Hippalgaonkar. (2024). Machine learning-assisted cross-domain prediction of ionic conductivity in sodium and lithium-based superionic conductors using facile descriptors. arXiv:2003.04922.
- [21] Seong-Hoon Jang, Randy Jalem, Yoshitaka Tateyama. (2024). Predicting room-temperature conductivity of Na-ion super ionic conductors with the minimal number of easily-accessible descriptors. arXiv:2404.01627.
- [22] Zirui Zhao, Xiaoke Wang, Si Wu. (2024). Deep learning-driven evaluation and prediction of ion-doped NASICON materials for enhanced solid-state battery performance. arXiv:2409.02952.
- [23] Hao Wu, Cameron Hargreaves, Arpit Mishra. (2024). Machine Learning for Electrode Materials: Property Prediction via Composition. arXiv:2603.07805.
- [24] Vidushi Sharma, Maxwell Giammona, Dmitry Zubarev. (2024). Formulation Graphs for Mapping Structure-Composition of Battery Electrolytes to Device Performance. arXiv:2307.03811.
- [25] Vidushi Sharma, Andy Tek, Khanh Nguyen. (2024). Improving Electrolyte Performance for Target Cathode Loading Using Interpretable Data-Driven Approach. arXiv:2409.01989.
- [26] Zhuohan Li, KyuJung Jun, Bowen Deng. (2024). Hierarchical high-throughput screening of alkaline-stable lithium-ion conductors combining machine learning and first-principles calculations. arXiv:2511.20964.
- [27] Wenjie Chen, Zichang Lin, Xinxin Zhang. (2024). AI-Driven Accelerated Discovery of Intercalation-Type Cathode Materials for Magnesium Batteries. arXiv:2412.11032.

- [28] Jiyeon Kim, Donggeon Lee, Dongwoo Lee. (2024). Machine Learning Prediction Models for Solid Electrolytes based on Lattice Dynamics Properties. arXiv:2404.13858.
- [29] Seungpyo Kang, Minseon Kim, Kyoungmin Min. (2024). Machine Learning-Aided Discovery of Superionic Solid-State Electrolyte for LiIon Batteries. arXiv:2202.06763.
- [30] Huican Mao, Xiang Zhu, Guangmao Li. (2024). A family of air-stable chalcogenide solid electrolytes in Li<sub>2</sub>BMQ<sub>4</sub> (B = Ca, Sr and Ba; M = Si, Ge and Sn; Q = O, S and Se) systems. arXiv:2405.03466.
- [31] Jonas Böhm, Aurélie Champagne. (2024). Predicting Crystal Structures and Ionic Conductivities in Li<sub>3</sub>YCl<sub>6</sub>–xBr<sub>x</sub> Halide Solid Electrolytes Using a Fine-Tuned Machine Learning Interatomic Potential. arXiv:2510.09861.
- [32] Xiuying Zhang, Jun Zhou, Jing Lu. (2024). Interpretable learning of voltage for electrode design of multivalent metal-ion batteries. arXiv:2201.04493.
- [33] Hongwei Du, Jian Hui, Lanting Zhang. (2024). Benchmarking Universal Machine Learning Interatomic Potentials for Solid-State Electrolytes: A Systematic Study on Energy, Forces, and Lithium-Ion Diffusivity. arXiv:2502.09970.
- [34] Xiaoqing Liu, Xinyu Yu, Yangshuai Wang. (2024). An AI-Ready Fine-Tuning Framework for Accurate Machine-Learning Interatomic Potentials in Solid–Solid Battery Interfaces. arXiv:2601.17847.
- [35] Jingchen Lian, Xiao Fu, Xuhe Gong. (2024). High-Throughput NEB for Li-Ion Conductor Discovery via Fine-Tuned CHGNet Potential. arXiv:2507.02334.
- [36] Selva Chandrasekaran Selvaraj, Volodymyr Koverga, Anh T. Ngo. (2024). Exploring Li-ion Transport Properties of Li<sub>3</sub>TiCl<sub>6</sub>: A Machine Learning Molecular Dynamics Study. arXiv:2403.01077.
- [37] Aqshat Seth, Rutvij Pankaj Kulkarni, Sai Gautam Gopalakrishnan. (2024). Investigating Ionic Diffusivity in Amorphous Solid Electrolytes using Machine Learned Interatomic Potentials. arXiv:2409.06242.
- [38] Yun An. (2024). Corner-Sharing PS<sub>4</sub>-BS<sub>4</sub> Modes Facilitate Fast Ion Conduction in Lithium Thioborophosphate Iodide Glassy Solid Electrolytes. arXiv:2507.08215.
- [39] Ankit Agrawal, Yierpan Aierken, Meiling Sun. (2024). Harnessing Structural and Dynamic Heterogeneity to Direct Ion Transport in Plastic Crystal–Polymer Composite Solid-Ion Conductors. arXiv:2009.04888.
- [40] Ke Li, Jitai Yang, Yu Zhai. (2024). Disentangling Cation-Polyanion Coupling in Solid Electrolytes: Which Anion Motion Dominates Cation Transport?. arXiv:2501.02440.
- [41] Qiye Guan, Kaiyang Wang, Jingjie Yeo. (2024). Unlocking the Ion Migration in Solid-State Electrolytes via Path Entropy. arXiv:2412.07115.
- [42] Shuo Tao, Qiang Zhu. (2026). Accelerated Inorganic Electrides Discovery by Generative Models and Hierarchical Screening. arXiv:2601.21077.
- [43] Joy Datta, Amruth Nadimpally, Nikhil Koratkar. (2024). Generative AI for Discovering Porous Oxide Materials for Next-Generation Energy Storage. arXiv:2410.06433.

- [44] Tri Minh Nguyen, Sherif Abdulkader Tawfik, Truyen Tran. (2024). Efficient Symmetry-Aware Materials Generation via Hierarchical Generative Flow Networks. arXiv:2411.04323.
- [45] Zhenze Yang, Yifan Wu, Xu Han. (2025). A Unified Predictive and Generative Solution for Liquid Electrolyte Formulation. arXiv:2504.18728.
- [46] Jiangyu Chen, Huawei Zhou, Zhou Zhang. (2024). Scientific Discovery under Imperfect Evaluators: Diversity-Aware Neuro-Symbolic Search for Solvent Design. arXiv:2603.03686.
- [47] Cheng-Man Wang, Chao-Hsiang Hsu, Jing-Sen Yang. (2024). Ab initio Study on Lithium Anode Interface Instability and Stabilization of Superionic Li<sub>3</sub>InCl<sub>6</sub> and Li<sub>6</sub>PS<sub>5</sub>Cl Solid Electrolytes. arXiv:2412.02433.
- [48] Mona Abdelgaid, Oliver Hvidsten, Théo Sombret. (2024). Identification of Solid-Electrolyte Interphase Species by Joint Characterization of Li-ion Battery Chemistry by Mass Spectrometry and Electro-Chemical Reaction Networks. arXiv:2602.17722.
- [49] Gracie Chaney, Andrey Golov, Ambroise van Roekeghem. (2024). Two-step growth mechanism of the solid electrolyte interphase in argyrodite/Li-metal contacts. arXiv:2402.04095.
- [50] Qiye Guan, Yongqing Cai. (2024). Unraveling Lithium Dynamics in Solid Electrolyte Interphase: From Graph Contrastive Learning to Transport Pathways. arXiv:2603.02284.
- [51] Ying Zhao, Runzi Wang, Emilio Martínez-Pañeda. (2024). A phase field electro-chemo-mechanical formulation for predicting void evolution at the Li-electrolyte interface in all-solid-state batteries. arXiv:2206.14753.
- [52] Ansgar Lowack. (2026). An Analytical Model of Critical and Subcritical Alkali Metal Dendrite Growth in Ceramic Solid Electrolytes. arXiv:2603.20113.
- [53] Yongliang Ou, Lena Scholz, Sanath Keshav. (2025). Non-Arrhenius Li-ion transport and grain-size effects in argyrodite solid electrolytes. arXiv:2510.18630.
- [54] Changdeuck Bae. (2024). Interfacial breathing as a dynamic failure law in all-solid-state batteries: amplitude, phase lag and dual-timescale memory as design principles. arXiv:2604.24075.
- [55] Hyeongjun Koh, Eric Detsi, Eric A. Stach. (2024). Observation of Order and Disorder in Solid-Electrolyte Interphases of Lithium-Metal Anodes. arXiv:2505.03956.
- [56] Oriol Sans-Planell, Nikolay Kardjilov, Ingo Manke. (2024). ANISSA: Advanced Neutron Imaging for Solid-State batteries in Action. arXiv:2508.10400.
- [57] Zebin Li, Shimao Deng, Yijin Liu. (2024). Machine Learning Enabled Graph Analysis of Particulate Composites: Application to Solid-state Battery Cathodes. arXiv:2512.16085.
- [58] Alireza Ghafarollahi, Markus J. Buehler. (2025). Autonomous Inorganic Materials Discovery via Multi-Agent Physics-Aware Scientific Reasoning. arXiv:2508.02956.
- [59] Wenwen Li, Yuki Orimo, Nontawat Charoenphakdee. (2025). Lang2MLIP: End-to-End Language-to-Machine Learning Interatomic Potential Development with Autonomous Agentic Workflows. arXiv:2605.14527.
- [60] S. Shayan Mousavi Masouleh, Corey A. Sanz, Ryan P. Jansonius. (2024). Artificial Intelligence-Enabled Optimization of Battery-Grade Lithium Carbonate Production. arXiv:2402.07000.

- [61] Felix Thelen, Moonjoo Kim, Geovane Arruda de Oliveira. (2024). Autonomous scanning electrochemical cell microscopy enables rapid exploration of large compositionally complex material spaces. arXiv:2606.00779.
- [62] Fusataka Kuniyoshi, Kohei Makino, Jun Ozawa. (2024). Annotating and Extracting Synthesis Process of All-Solid-State Batteries from Scientific Literature. arXiv:2002.07339.
- [63] Guillaume Lambard. (2026). Beyond Structure: Revolutionising Materials Discovery via AI-Driven Synthesis Protocol-Property Relationships. arXiv:2605.00313.
- [64] Scott R. Broderick, Kaito Miyamoto, Krishna Rajan. (2024). High-Throughput Computation of Li-based Battery Material Databases: Chemistry-Processing-Property Relationships. arXiv:2203.08699.
- [65] Max C. Gallant, Matthew J. McDermott, Bryant Li. (2024). ReactCA: A Cellular Automaton for Predicting Phase Evolution in Solid-State Reactions. arXiv:2407.19124.
- [66] Antonino Curcio, Antonio Gianfranco Sabato, Marc Nuñez Eroles. (2024). Ultrafast crystallization and sintering of  $\text{Li}_{1.5}\text{Al}_{0.5}\text{Ge}_{1.5}(\text{PO}_4)_3$  glass and its impact on ion conduction. arXiv:2211.08225.
- [67] Thiago Reschützegger, Sarp Aykent, Gabriel Jacob Perin. (2026). EQUIVARIANT INTER-ATOMIC POTENTIALS WITHOUT TENSOR PRODUCTS. arXiv:2601.15492.
- [68] Dibakar Datta. (2024). Electro-Chemo-Mechanical Modeling of Multiscale Active Materials for Next-Generation Energy Storage: Opportunities and Challenges. arXiv:2309.01954.
- [69] Syed Mustafa Shah, Mohammed Lemaalem, Anh T. Ngo. (2024). Predictive Machine Learning Molecular Dynamics of SEI Formation in Concentrated LiTFSI and LiPF<sub>6</sub> Electrolytes for Lithium Metal Batteries. arXiv:2602.05141.
- [70] Junyu Jiao, Genming Lai, Liang Zhao. (2024). Self-healing mechanism of lithium in lithium metal batteries. arXiv:2106.10979.
- [71] Md Salman Rabbi Limon, Curtis Duffee, Zeeshan Ahmad. (2024). Constriction and contact impedance of ceramic solid electrolytes. arXiv:2501.00600.
- [72] Shafee Farzanian, Imtiaz Shozib, Nikhil Sivadas. (2024). Mechanical investigations of composite cathode degradation in all-solid-state-batteries. arXiv:2307.00145.
- [73] Yue Yin, Hai Xiao. (2024). Oxidation States in Solids from Data-Driven Paradigms. arXiv:2503.19425.